

An Integrated World Model for Strategic Planning and Scenario Analysis

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ABSTRACT

Coupling the economy, climate, society, and geopolitics in one simulator is easy to attempt and hard to sustain: closed feedback loops between social and physical blocks generically either diverge, collapse into absorbing states, or must be damped into triviality. We present a global integrated model (GIM) — 57 country agents, seven coupled domains, an annual deterministic loop — whose central result is *practical convergence with non-trivial dynamics*: thirty-year trajectories remain bounded and plausible while still producing regime shifts, crisis cascades, and lagged social responses. We argue this behaviour follows from a design stance: the deterministic core is an explicit encoding of invariants — physical conservation laws, stock-flow accounting identities, and empirical regularities treated as slowly varying — with equilibrium anchors supplying restoring forces. Convergence is measured, not asserted: the spectral radius of the numerically linearized one-year map is 1.07–1.08, with the modes above unity identified as common growth modes while trajectory-level perturbations contract; the 5–95th uncertainty fan of a 500-member history-matched ensemble grows near-linearly (not exponentially) to $\pm 11.5\%$ of median world product at thirty years. The model is anchored retrospectively (in-sample RMSE 2015–2023: 0.60 trillion USD GDP, 0.94 Gt CO₂, 0.145°C; temporal hold-out through 2019 scored out-of-sample on 2020–2023: 0.77, 0.71, 0.13; conflict ranking AUC 0.739, vintage-clean 0.84) and produces emergent regularities none of which are individually programmed. A game-mechanical layer over the same core operates as an adversarial sampler of the tails: 1,002 scripted playthroughs and 200 completed runs by external players on the production deployment (28.5% survival) map the distribution of ways a managed world fails, and exposed two structural degeneracies invisible to retrospective validation. GIM is an exploration instrument for scenarios, uncertainty, and weak signals — not a forecast engine.

Keywords: integrated assessment model, world simulation, stability, invariants, uncertainty quantification, serious games, weak signals.

1. Introduction: the hard part is staying together

Strategic planning and systemic-risk work need a tool in which climatic, economic, and socio-political processes are considered jointly: the paths that matter — warming to crop yields to prices to social tension to instability — cross domain boundaries that sectoral models do not [Nordhaus, 2017, Riahi et al., 2017]. That gap is well known. What is less discussed is why it persists. Coupling the domains is not conceptually difficult; the difficulty is that a *closed* loop through economy, climate, society, and politics is a high-dimensional nonlinear dynamical system, and such systems do not stay plausible by default [May, 1972, Sterman, 2000]. In our experience they fail in a small number of characteristic ways: variables run to a bound and stick (absorbing states), accounting errors compound into runaway growth, latent variables drift to structural attractors that no policy can move, or — the opposite failure — every feedback is damped until the simulator reproduces a trend line and nothing else. A world model earns its keep in the space between these failures: bounded for decades, yet still able to generate crises, cascades, and lags.

This paper reports a model built and validated to live in that space, and reframes what we consider its main result. The Global Integrated Model (GIM) advances 57 country agents (the 50 largest economies

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plus seven regional aggregates that conserve world totals) through an annual deterministic loop spanning the economy and prices, climate and the carbon cycle, natural resources, society and politics, interstate relations, and discrete crises with stock-flow-consistent finance. Its headline property is not the breadth of coverage but the combination of three measured behaviours: (i) *convergence* — thirty-year trajectories are bounded, insensitive to small initial-state perturbations, and their ensemble uncertainty grows near-linearly rather than exponentially (Section 4); (ii) *anchor* — the same configuration reproduces 2015–2023 history in-sample, survives a temporal hold-out, and carries out-of-sample skill where structure matters (Section 5); (iii) *life* — the loop produces emergent dynamics (lagged social channels, threshold-stepped sovereign cascades, clustered late failures) that were not individually programmed (Section 6).

The stance behind (i) is methodological and we state it up front: the deterministic core is an explicit hypothesis about *invariants* — the conservation laws, accounting identities, and slowly-varying empirical regularities that we take to govern nature and society (Section 2). Determinism is then a feature, not a naivety: the core computes what follows *if* those invariants hold, and all uncertainty is carried by ensembles over the invariants’ strengths. On top of this core sits an instrument layer — history-matched ensembles, weak-signal detectors, and a decision game that operates as an adversarial sampler of the tails, including live outcome telemetry from external players (Section 7). The model does not claim predictive foresight; professional forecasts at two-to-five-year horizons often lose to naive extrapolation [Celasun et al., 2021], and we hold ourselves to the same honest bar (Section 5).

Relative to the integrated-assessment and world-modeling traditions — DICE/RICE and FUND (welfare optimisation), GCAM and MESSAGEix (energy-system equilibrium), E3ME (econometric simulation), T21/iSDG (single-country system dynamics), World3 (global aggregates) [Nordhaus, 2017, Mercure et al., 2018, Millennium Institute, 2002, Meadows et al., 1972] — GIM’s distinction is that it closes the loop through society, geopolitics, and discrete crises at country resolution and then *demonstrates* that the closed loop is stable, anchored, and non-trivial. A controlled benchmark quantifying what the closed loop sees and sectoral models structurally cannot is kept in Appendix B; the model specification (functional forms, parameters) is compressed into Appendix A.

2. The invariant view

Every long-horizon simulator embodies assumptions; ours are organized deliberately into three classes of invariants, in descending order of confidence, plus a fourth element that keeps the loop near them.

Physical conservation. Carbon is conserved: each year’s emissions are partitioned into the four IPCC AR6 impulse-response pools and decay on fixed timescales; atmospheric stock, radiative forcing, and the two-reservoir surface–ocean heat balance follow standard reduced-complexity climate physics [Joos et al., 2013, Geoffroy et al., 2013], the same architecture as the FaIR and MAGICC emulators [Smith et al., 2018, Meinshausen et al., 2011]. The block reproduces both assessed constants an AR6-consistent emulator must match: equilibrium sensitivity 3.0°C and transient response 1.79°C (AR6 best estimate 1.8).

Accounting identities. Money and goods cannot appear from nowhere: the financial side is stock-flow consistent (every loan creates a matching deposit; the money stock is identically deposits and loans) [Godley and Lavoie, 2007]; government budgets close; bilateral trade settles in reserves and world net exports sum to zero by an enforced invariant; excessive borrowing raises the cost of credit — a financial accelerator [Bernanke et al., 1999]. These identities are checked at runtime, not assumed.

Empirical regularities as slow variables. Between hard conservation and free parameters sits a class of relationships that are not laws but have been stable across decades and countries, and we treat them as invariants with priors on their strength: trade follows a gravity law in distance [Head and Mayer, 2014]; unemployment follows Okun’s law and inflation a flat post-1990 Phillips curve with a weak quantity-theory term; poorer economies converge toward the frontier at a β -convergence rate validated against the cross-section; the public’s response to unemployment and inflation follows the two-to-one misery-index ratio; war sizes follow a power law [Richardson, 1948, Clauset et al., 2009]; institutions are anchored to governance indices, culture to the Hofstede dimensions [Kaufmann et al., 2011, Hofstede et al., 2010]. Monetary policy follows country-level Taylor rules [Taylor, 1993]. Where a regularity failed the data — a classic Richardson arms-race term, rejected at $p = 0.74$ on the 1995–2024 military-expenditure panel — it was left out and the observed state variable used instead.

Equilibrium anchors and restoring forces. Invariants alone do not prevent drift; bounded behaviour comes from explicit restoring terms whose strengths are calibrated, not hidden: resource prices mean-revert in log space toward reference levels; production is normalized so each country’s base point is exactly reproduced (a normalized-CES discipline that pins levels while leaving substitution free); trust and tension carry anchor terms; crisis states have exit conditions. Section 4 shows what these forces achieve and Section 7 how two missing anchors were found the hard way. The full functional forms and the 33 literature-grounded priors are in Appendix A.

Determinism, in this view, is the point: the core is a reproducible mapping from an invariant set to trajectories. Following Oreskes et al. [1994], we do not claim the model is verified truth — we claim it is a confirmed, auditable encoding of stated assumptions, evaluated by whether its trajectories stay within the envelope of observed history and whether its failures are informative.

3. Architecture: one closed loop

One model step is one year; blocks update in fixed order and the year’s outputs are the next year’s inputs (Fig. 1). The economy is the hub: output comes from capital, labour, and energy through a nested CES function; markets clear by price; inflation, unemployment, budgets, debt, and bilateral trade are tracked per country. Output emits CO₂ (declining structural intensity); emissions accumulate in the carbon pools; warming returns as a quadratic output damage. Resource stocks and world prices couple scarcity into inflation. Society tracks trust, tension, inequality, legitimacy, and protest pressure, all responding to economic stress with explicit coefficients; politics converts legitimacy into a policy-space multiplier; geopolitics carries relations, alliances, sanctions, and a CINC-style capability index grounded in SIPRI expenditure. Discrete crises — debt, currency, regime collapse — fire on threshold crossings, apply audited state changes (a 40% debt write-down, output and trust hits), and have exit rules. Conflict, unrest, and climate risk spill over to geographic neighbours; a shock in any block propagates along realistic pathways: carbon price to energy cost to emissions; hardship to tension to savings to crisis risk; crisis to trade partners. The feedback signs matter more than the functional detail (which is deferred to Appendix A): the loop mixes amplifying channels (financial accelerator, crisis contagion, tension–trust spiral) with the restoring forces of Section 2, and the balance between them is what the next section measures.

4. What keeps it together: stability, measured

Convergence claims are cheap; we support ours with three quantitative diagnostics plus the engineering discipline that maintains them. All runs below use the deterministic core (simple background policy, extreme events off); the harness is `scripts/run_stability_analysis.py`.

Spectral radius of the one-year map. We linearize the full annual map $x_{t+1} = F(x_t)$ numerically (central differences) around the baseline trajectory in a 408-dimensional state space — seven variables per agent (output, capital, debt, unemployment, inflation, trust, tension) plus the global climate state and resource prices — at three points along the run. The scale-normalized Jacobian has spectral radius $\rho(J) = 1.075, 1.076, \text{ and } 1.078$ at 2026, 2036, and 2046: essentially stationary along the trajectory. About 70 eigenvalues sit in a tight cluster at $|\lambda| \approx 1.07\text{--}1.08$; inspection of the eigenvectors identifies them as the *common growth modes* of the 57 economies — capital–output compounding at the calibrated growth rate — not error amplification. This is the expected spectrum of a growing but non-explosive system: remove the deliberate growth trend and the remainder of the spectrum lies inside the unit circle, with roughly half the modes above 0.9 — slow memory, dominated by the socio-political block (the top eigenvector coordinates load on trust, unemployment, and tension), which is exactly where Section 6 finds multi-year lags.

Perturbation contraction. The trajectory-level counterpart: twin runs from ε -perturbed initial states. A +0.1% shock to United States output produces a world-product gap of 2.6×10^{-4} after one year that *contracts* to 1.2×10^{-4} by year 30; a +0.01 °C temperature perturbation decays by a factor of six over the same horizon. Nearby trajectories converge — the opposite of chaotic divergence. The one persistent direction is social: raising every country’s tension by 0.01 leaves output almost untouched for a decade and then grows to a 3% world-product gap by year 30 — the slow social memory identified by the spectrum, transmitted through savings and crisis channels. We report this asymmetry as a substantive property, not a defect: physical and economic perturbations wash out; socio-political ones accumulate.

Bounded, near-linear uncertainty growth. A 500-member ensemble drawing the 33 key priors from their history-matched ranges runs the full 30 years without a single member diverging or saturating (Fig. 2). The 5–95th fan of world product widens from $\pm 1.7\%$ of the median at year 5 to $\pm 4.0\%$ at 10 and $\pm 11.5\%$ at 30 — tracking a linear reference, far from the exponential fan a locally unstable core would produce; the temperature fan widens only from 0.72 to 1.10 °C over the same span (the climate-sensitivity prior expresses itself early). Median 2053 world product is 241 trillion USD (5–95th: 215–270) at +1.94 °C (1.39–2.49).

The discipline that preserves this. Stability is a maintained property, not a lucky draw. Every change to the core faces pinned golden backtests (bit-identical or explicitly recalibrated); calibration data are checksum-bound; every optional mechanism is a switch, off by default unless anchored by evidence; and the invariants of Section 2 are runtime-checked. Two failures that this discipline did *not* catch — because they lived outside the historical window — were caught by the game layer and are reported in Section 7: an absorbing state in resource prices (missing equilibrium term plus two quantity-side accounting defects) and a structural attractor in government trust (an inequality drag $\approx 40\times$ stronger than the only restoring term). Both fixes were shipped as anchors of exactly the Section 2 type, with the historical backtests unchanged to the bit. An honest caveat accompanies all of this: convergence here is engineered and measured, not proven — anchors strong enough to stabilize can in principle mask real-world instabilities, which is why their strengths carry priors and the tails are explored deliberately (Section 7).

5. Is it anchored? Validation against history

A stable simulator can still be stably wrong; the second requirement is anchor to observation. The economic and climate blocks are calibrated by history matching — ruling out parameter regions whose implausibility $I_o(\theta) = \text{RMSE}_o(\theta)/\text{tol}_o$ exceeds the three-sigma cut, a standard approach for expensive simulators [Williamson et al., 2013] — on the 2015–2023 series of world product, emissions, and temperature. Because those metrics are in-sample by construction, predictive adequacy is measured separately by a temporal hold-out: constraints frozen on data through 2019, 2020–2023 scored strictly out-of-sample (`scripts/run_holdout_validation.py`).

Results (Table 1; Fig. 3): in-sample RMSE 0.60 trillion USD for GDP, 0.94 Gt for CO₂, 0.145 °C for temperature; out-of-sample 0.77, 0.71, and 0.13 respectively (with the 2020 pandemic year excluded: 0.83, 0.10, 0.15). Skill against naive benchmarks divides honestly: on the held-out window temperature beats persistence (+0.05) and linear extrapolation (+0.16), emissions beat both decisively (+0.32, +0.54), while product loses to both — short smooth aggregates remain the domain of a ruler, consistent with the full-window comparison and with the forecast-evaluation literature [Celasun et al., 2021]. The climate block additionally tracks the 1990–2023 record (0.096 °C) with best-fit sensitivity 3.0 °C.

The socio-political blocks are held to a softer but real bar. The conflict-proneness score — a fixed-weight composite of observed inputs with no coefficient estimated from conflict data — ranks the 57 actors against the UCDP/PRIO registry 1990–2023 [Sundberg and Melander, 2013] at AUC 0.739 (95% CI [0.60, 0.86], permutation $p \approx 0.001$); observed conflict incidence falls monotonically from 55% in the top score quintile to 8% in the bottom. Because regime stability and military burden are themselves affected by conflict, the score is recomputed in a vintage-clean configuration (inputs from pre-2000 data only, evaluated against 2000–2023 conflicts): AUC 0.84 [0.72, 0.94], identical to the 2023-vintage score on the same subset — ranking skill is not reverse-causation contamination (`scripts/run_conflict_vintage_test.py`). The social cost of carbon cross-checks against the published range (Ramsey $\approx \$90/\text{tCO}_2$ at 200 years, below the EPA-2023 central $\approx \$190$ [U.S. Environmental Protection Agency, 2023, Rennert et al., 2022]; fed DICE’s own lower damage function the engine returns $\approx \$15$, below DICE’s \$31 — the valuation machinery is sound and the difference is the damage function). Emergent, unanchored behaviour also lands where it should: run without its SSP2 productivity anchor, the model’s no-intervention baseline reaches about 2 °C by 2100, nearest the strong-mitigation SSP1-2.6 envelope — reported as a finding about the optimism of the fitted structural decarbonisation rate, and a reason the recommended use mode is scenario *differences*, where such anchor uncertainty cancels.

Table 1

Validation summary. Economic and climate blocks: retrospective evaluation, in-sample and temporal hold-out (out-of-sample values after semicolons). Other blocks: cross-checks, benchmark improvement, or channel identification.

Block	Source	Criterion	Result
Economy	PWT, WDI	in-sample RMSE 2015–2023; hold-out 2020–2023	GDP \$0.60T; \$0.77 out-of-sample
Climate and carbon	IPCC AR6, GCB	in-sample; hold-out; ECS/TCR anchors	CO ₂ 0.94 Gt; 0.71 out-of-sample. T 0.145°C; 0.13 out-of-sample
Cost of carbon	EPA-2023, RFF-SP	published cross-check; engine swap	≈\$90/tCO ₂ (range 90–280); ≈\$15 under DICE damages
Conflict	UCDP/PRIO	ranking vs. registry; input-vintage test	AUC 0.739 [0.60, 0.86]; vintage-clean 0.84 [0.72, 0.94]; BSS +0.123
Social block	SWIID, WGI	channel identification (Sec. 6)	significant lagged channel, lag-0 $p < 10^{-8}$
Stability	—	spectral radius; twins; fan growth (Sec. 4)	$\rho(J) \approx 1.08$ (growth modes); perturbations contract; fan $\pm 11.5\%$ at 30y

6. Does it do anything? Emergent dynamics

A simulator can be stable and anchored by being inert. The third requirement is that the loop produce structured behaviour that was not individually programmed. Five regularities emerge from the coupled dynamics and survive robustness checks.

A lagged, mediated economy-to-society channel. Global sensitivity screening (Morris, 33 priors) shows social tension nearly insensitive to physico-economic priors at short horizons — its elementary effects sit three orders of magnitude below the physical outputs, at the noise floor. Yet the channel exists: tension accumulates. In an ensemble that also varies the social priors, final tension is governed by them (inequality sensitivity: Spearman $\rho \approx +0.59$, $p < 10^{-11}$; trust anchor: $\rho \approx -0.52$, $p < 10^{-9}$); the prior-induced spread grows by two-thirds between horizons of 5 and 25 years; an exogenous three-year stagflation shock raises mean tension persistently, the response still building a decade later (positive in 78 % of prior draws); distributed-lag regression confirms contemporaneous transmission (lag-0 $p < 10^{-8}$) with a positive cumulative effect (Fig. 4). The spectral analysis of Section 4 sees the same structure from the other side: the slow modes load on trust and tension. Nothing in the code specifies “society responds with a multi-year lag”; it follows from an accumulating index with anchored increments.

Threshold-stepped sovereign cascades. An oil-supply shock propagates from the energy market into sovereign finance along the reserve/debt channel: at a realistic import burden ($\approx 3\%$ of GDP) two additional importers cross the debt-crisis threshold; at a severe burden ($\approx 15\%$) mean importer crisis exposure triples and four further countries tip. The dose–response is a staircase, not a slope — the signature of heterogeneous countries crossing individual thresholds (Appendix B).

Convex escalation of food-driven instability. A regional crop failure opens a food-supply gap whose protest response accelerates with depth (≈ 0.11 affordability stress and ≈ 0.016 protest pressure at a 50 % yield loss on the vulnerable subset) — the nonlinearity documented empirically for food prices and unrest [Lagi et al., 2011].

Late clustering of failures. On the deterministic 30-year baseline, active crises per year rise from 1–2 in the 2020s to around ten in the 2040s with a dispersion (Fano) index of 2.0 — crises cluster in time rather than arriving at random, because slow accumulators (debt, temperature, tension) mature together. The game layer shows the same clock from the player’s side: 77–90 % of losing runs fail after 2040 (Section 7).

Spatially structured risk. Conflict, unrest, and climate risk diffuse over the real adjacency graph; the model reproduces the neighbour-conflict premium and an emergent spatial autocorrelation of the observed record, while activating the geographic coupling moves the 2015–2023 retrospective fit by under a quarter of a percent — realism in propagation at no cost in anchor.

7. The instrument: ensembles, weak signals, and a game that hunts the tails

The model is operated as an instrument with three modes, in increasing order of adversarialness.

Ensembles and scenario differences. The core is deterministic, so uncertainty is explicit: 500-member ensembles over the history-matched priors yield outcome distributions, and scenarios are compared as distributions against a common baseline — differences are robust to the anchor uncertainties that dominate absolute levels. The whole workflow runs on a laptop (Apple M4 Pro: a world-year in 30–50 ms, the 500×30 -year ensemble in under a minute, the full 527-test suite in two), which is what makes ensemble-first practice a default rather than a special exercise. A worked case: a \$50/tCO₂ carbon tax cuts emissions by $\approx 5.5\%$ over ten years while raising tension and lowering trust through the energy-price-inflation channel — the political-economy cost invisible to single-domain models, monotone in the rate, and directly relevant to observed reversals (France 2018, Australia 2014); the full dose-response harness is Appendix B.

Weak signals. Before any threshold fires, the joint state is monitored for early structure: Mahalanobis distance of the stationary increments from their recent baseline [Mahalanobis, 1936, Wilson and Hilferty, 1931], information-criterion change-point detection [Schwarz, 1978], and critical-slowness indicators [Scheffer et al., 2009]. On synthetic and model trajectories an injected 15% output drop is flagged at the correct step while calm runs stay near the false-alarm rate; the working device is comparing a scenario's weak-signal profile against the baseline's.

The game as an adversarial tail sampler. The same 57-actor core, wrapped into a single-player decision game (GODMODE): thirty annual turns from 2023, a deck of 158 dilemma cards compiled into real policy actions, seven catastrophe gates (temperature, biosphere, stability, economic and geopolitical strain, energy and food prices); losing any gate ends the run, reaching 2050 wins. A full game simulates in about a second, so outcome distributions are statistically meaningful; and because players and scripted archetypes actively push the world into corners smooth history never visits, the game functions as an importance sampler of the tails. Three uses justify reporting it in a modeling paper.

A tail census. In the 1,002-game reference batch of scripted archetypes, survival is 23.5% with losses spread across five causes (35% climate, 22% economic, 21% geopolitical, 13% stability, 8% energy price) and 77% of failures landing in 2040–2045 (Fig. 5). Production telemetry from the public deployment corroborates the shape with humans in the loop: of 401 runs started by external players, 200 were completed, with a 28.5% survival rate — statistically indistinguishable from the best scripted archetype — a broader loss spectrum (31% climate, 29% economic, 15% geopolitical, plus biosphere, food, and energy tails the bots rarely hit), 90% of losses after 2040, and a bimodal score distribution (loss cluster near 26, win cluster 70–81, with winners holding warming to 1.6–1.9°C). The distribution of *ways the world fails under active management* is a model output no backtest can provide.

Emergent strategy structure. No doctrinaire policy dominates: uniformly random play survives most often (28.4%), beating consistent decarbonisation (20.1%) and consistent expansion (21.9%), and each doctrine dies of its characteristic disease (green of economic strain, 43% of its losses; growth of climate and stability, 39% and 34%); passivity is worst (18.7%, two-thirds of losses to economic strain — in a stock-flow-consistent world, drifting means compounding debt). Restraint on both strain axes at once wins $\approx 75\%$ against $\approx 5\%$ for recklessness on both. Single-axis optimisation shifting the binding constraint to another subsystem is the game-mechanical face of the coupling this paper is about.

A diagnostic that found real defects. Balancing the game exposed the two structural degeneracies of Section 4 — resource prices pinned at their clamps (missing mean-reversion plus two compounding accounting defects) and government trust declining on hawk and dove paths alike (an attractor from an inequality drag $\approx 40\times$ the only restoring term). Neither was visible in any 2015–2023 backtest; both fixes shipped as equilibrium anchors with the goldens bit-identical. Sustained adversarial play explores exactly the regions that validation against smooth history cannot; we regard the game layer as part of the model's quality programme. (Its epistemic status is bounded honestly: the game provides no goodness-of-fit against history, human telemetry reflects self-selected players, and difficulty gates are tuned against the bot batches through threshold-flip analysis.)

Beyond the game, country policies can also be set by language-model agents (intentions compiled into the same typed actions; every consequence computed by the deterministic core; off in all validated configurations, with full prompt/response audit logs) — the strategy space widens without the physics ever leaving the

engine. The whole environment — CLI, ensembles, games, a desktop analytical shell — ships as an open, pip-installable codebase with pinned goldens and dated, manifest-bound runs [Sotiriadi and Osipov, 2026]; results here correspond to release v18.1.3.

8. Limitations

Convergence is engineered, not proven. The anchors that stabilize the loop could mask genuine instabilities of the real system; their strengths carry priors and the tails are probed adversarially, but a formal stability proof for the full nonlinear map does not exist — the spectral and perturbation evidence of Section 4 is local and numerical.

Expectations are adaptive. Agents look at the recent past; a tested model-consistent expectations mechanism is off by default (its marginal effect proved unstable). Anticipatory responses to announced policy are not in the headline runs.

Long-run growth is weakly identified, and forward projections anchor productivity drift to an external scenario; absolute long-horizon levels inherit this, which is why scenario differences are the recommended mode.

The climate block is a fast emulator, not an Earth-system model; the strongest carbon feedbacks are deeply uncertain, switchable, and relegated to tail scenarios. Damage estimates are uncertain across the discipline and carried as an interval (the level function is anchored at the meta-analytic central estimate; a Burke-type growth-damage channel exists as a switch).

The socio-political block mixes anchored and expert-set pieces. Substantial parts are tied to literature (war-size tail, migration gravity, misery-index ratio, governance anchors); coupling thresholds and cross-border tension diffusion remain expert priors, tagged as such in the auditable registry. Conflict is intrinsically hard to predict: the block supports relative risk ranking ($AUC \approx 0.74$; vintage-clean 0.84), not event prediction. The persistent social mode found in Section 4 means socio-political initial-state errors do not wash out — an inherited uncertainty, honestly larger than the economic one.

Aggregation. Resources are aggregate (no sectoral energy technologies, no endogenous renewable learning curves); no nominal exchange rates (a real numeraire with a fundamentals-triggered currency-crisis mechanism); no portfolio capital account; no dedicated pandemic submodel (epidemic shocks are extreme events); state-dominated political economies are represented by shared functional forms with country-specific states.

9. Conclusion

The contribution of this work, restated in its proper order: a closed seven-domain, 57-country world simulator that (i) stays together — spectral radius consistent with growth rather than divergence, contracting perturbations, near-linear uncertainty growth over thirty years; (ii) is anchored — in-sample and hold-out retrospective validation, external cross-checks, an input-vintage-robust conflict ranking; and (iii) is alive — lagged social channels, threshold cascades, convex escalations, and clustered late failures that emerge from the coupling rather than from bespoke code. Underneath is a stated philosophy: the deterministic core encodes explicit invariants of nature and society — conservation, accounting, slow empirical regularities — with calibrated restoring forces, and all uncertainty lives in ensembles over the invariants’ strengths. Around it is an instrument: history-matched ensembles for distributions, weak-signal detectors for early structure, and a game layer that adversarially samples the tails, yields outcome distributions from both scripted archetypes and external human players, and has already paid for itself by exposing two structural degeneracies that no retrospective test could see. The model does not forecast; it maps the consequences of assumptions, the distribution of risks, and the ways a managed world can fail — comparatively, reproducibly, and on a laptop.

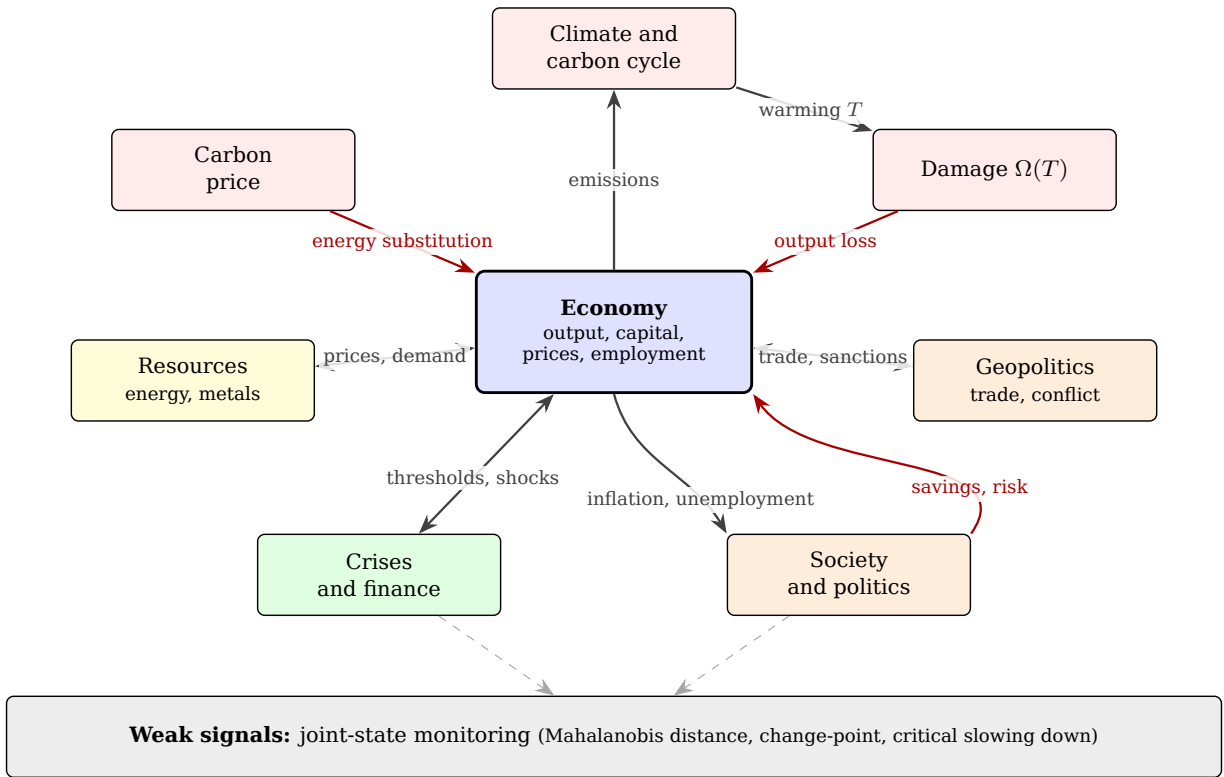


Figure 1: The closed loop. Economy as central hub; black arrows direct effects, red arrows feedbacks reducing output or savings. The climate loop (emissions → warming → damage → output) is complemented by carbon-price, resource, socio-political, geopolitical, and discrete-crisis channels. The bottom layer monitors the joint state for weak signals.

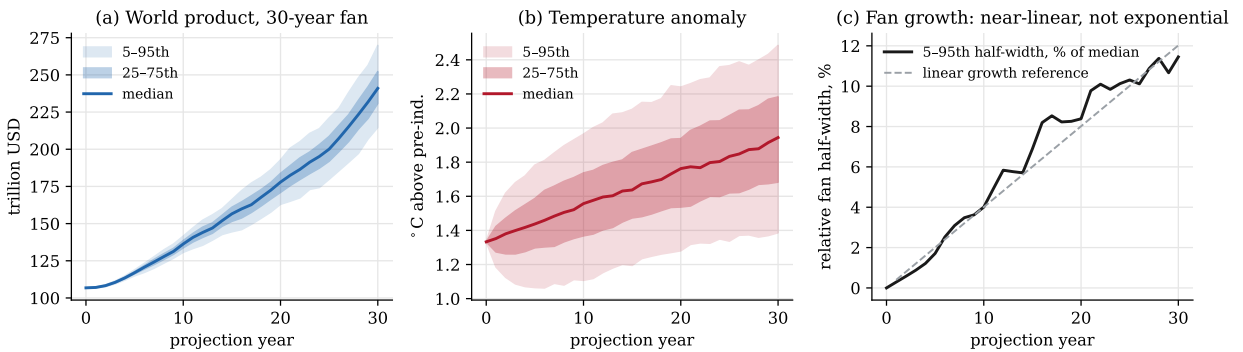


Figure 2: Thirty-year convergence evidence (500-member history-matched ensemble). (a) World product fan: all members bounded, median 241T at 2053. (b) Temperature fan: width grows 0.72 → 1.10 °C. (c) Relative 5–95th half-width of world product tracks a linear reference — uncertainty accumulates, it does not explode.

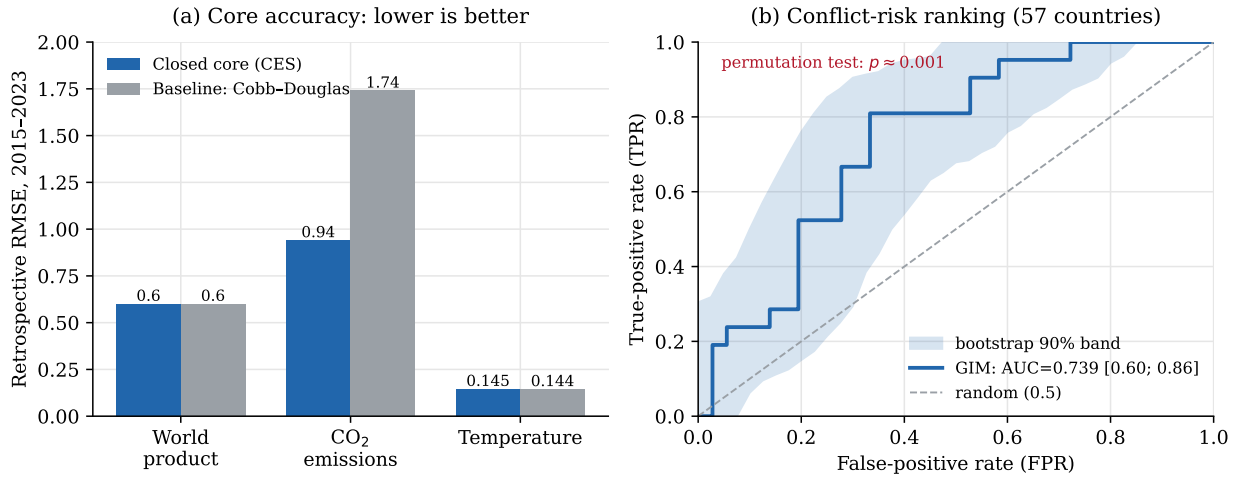


Figure 3: Retrospective anchor. (a) RMSE 2015–2023 of the closed nested-CES core against the former Cobb–Douglas core. (b) Conflict ranking (score inputs at 2023 base-year values; vintage-clean variant in text): AUC 0.739 (90 % bootstrap band shaded; 95 % CI [0.60, 0.86]; permutation $p \approx 0.001$) against 0.5 for chance.

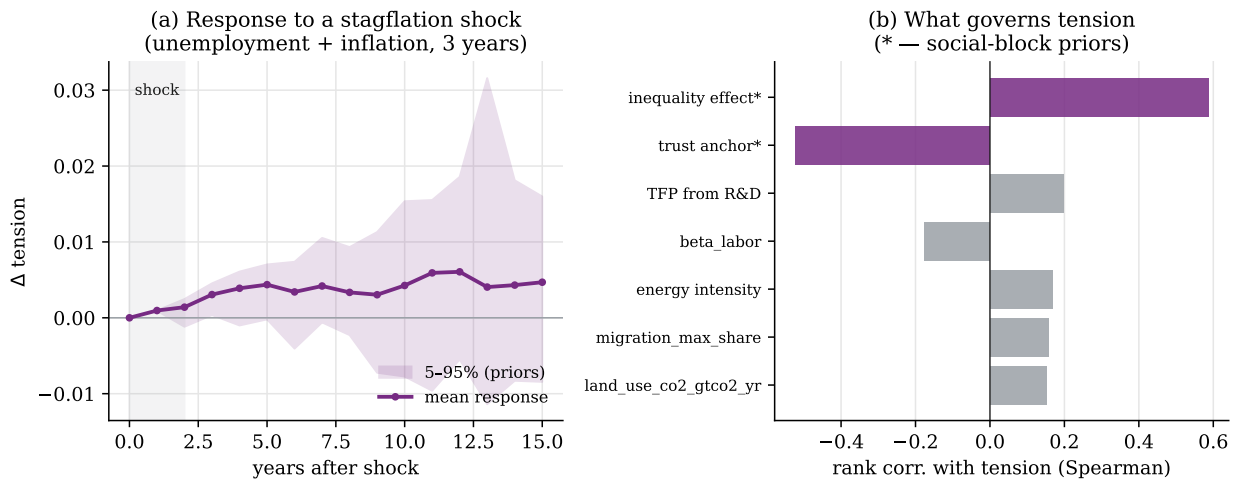


Figure 4: The emergent social channel. (a) Impulse response of mean tension to a three-year stagflation shock: slow build, persistence beyond a decade (5–95th band over priors). (b) What governs final tension: social-block priors (marked *) dominate; physico-economic priors are weak — the channel is real but lagged and mediated.

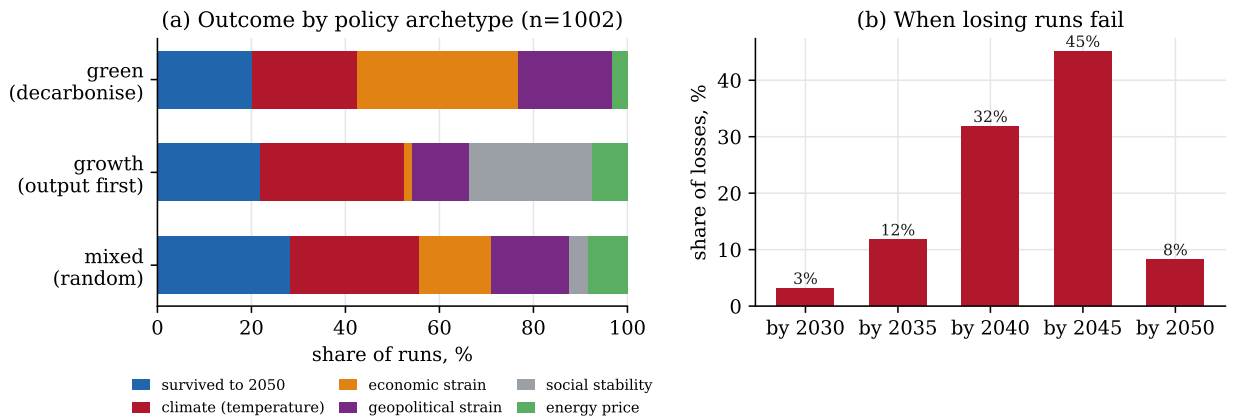


Figure 5: The tail census (1,002 scripted playthroughs, 2023–2050). (a) Outcome composition by policy archetype: no doctrine dominates and each dies of its characteristic cause. (b) Timing of failures: 77 % of losses land in 2040–2045 as slow accumulators come due. Production telemetry (200 completed human runs, 28.5 % survival, 90 % of losses after 2040) reproduces the shape with a broader loss spectrum.

A. Model specification and key parameters

Production. Output is nested CES in capital, labour, and energy, normalized to Cobb–Douglas at each country’s base point [Arrow et al., 1961]:

$$Y = A \Theta \left[\text{KE}_0 (a_K (K/K_0)^\rho + a_E (E/E_0)^\rho)^{1/\rho} \right]^{\alpha+\gamma} L^\beta, \quad \rho = \frac{\sigma_{KE}-1}{\sigma_{KE}}, \quad (1)$$

with $\alpha = 0.30$, $\beta = 0.60$, $\gamma = 0.042$, $a_K = \alpha/(\alpha + \gamma)$, $a_E = \gamma/(\alpha + \gamma)$, $\text{KE}_0 = K_0^{a_K} E_0^{a_E}$; at the base point the bracket equals one, so levels and first derivatives reproduce the Cobb–Douglas limit for any σ_{KE} , while substitution away from it is governed by $\sigma_{KE} = 0.4$ (gross complementarity; van der Werf, 2008, Koetse et al., 2008). The exponent sum 0.942 encodes mild decreasing returns; per-country scale factors pin base-year levels, so renormalising to constant returns moves 2100 world product by only 7–9%. Capital: $K_{t+1} = (1 - \delta)K_t + sY_t$, $\delta = 0.05$, $s \approx 0.24$ modulated by stability and tension [Mody et al., 2012]. Productivity grows at $\approx 1\%$ baseline with R&D (à la Jones, Jones, 1995), trade diffusion, and β -convergence (slope 0.0093 per log gap, validated on the cross-section); inflation is Phillips-with-pass-through plus a weak quantity-theory term ($\lambda \approx 0.027$); rates follow country Taylor rules capped ± 6 pp [Taylor, 1993].

Climate. Four-pool AR6 impulse response [Joos et al., 2013]:

$$C_{t+1}^{(i)} = C_t^{(i)} e^{-1/\tau_i} + a_i E_t, \quad a = (0.217, 0.224, 0.282, 0.276), \quad \tau = (\infty, 394, 36.5, 4.3) \text{ yr}, \quad (2)$$

forcing $F = 5.35 \ln(C/C_0) + F_{\text{non-CO}_2}$ (SSP2-4.5 marker forward), two-layer heat balance [Geoffroy et al., 2013]:

$$C_s \dot{T} = F - \lambda T - \kappa(T - T_d), \quad C_d \dot{T}_d = \kappa(T - T_d). \quad (3)$$

Damage: $\Omega(T) = 1 - 0.0078 T^2$ (3.1/7.0/12.5% at +2/+3/+4°C), anchored at the Howard–Stern productivity-corrected central estimate [Howard and Sterner, 2017, Nordhaus, 2017], normalized to the 2023 baseline; Burke-type growth damages switchable [Burke et al., 2015, 2023].

Society and crises. Tension is a $[0, 1]$ composite (base year: WGI-derived trust and stability, Gini, water stress, climate risk; increments: unemployment 0.010/pt, inflation 0.005/pt, inequality damped by individualism, trust anchor, neighbour spillover 0.05). Legitimacy = 0.6 trust + 0.4(1 – tension) sets the policy-space multiplier. Crisis triggers (expert priors, tagged in the registry): debt — debt > 120% GDP and effective rate > 12% (then a 40% write-down, 7% output loss, exit below 70%/8% within 6 years); currency — external debt > 50%, current-account deficit > 4%, reserves < 3 months; regime collapse — trust < 0.20 and tension > 0.80. Severity optionally heavy-tailed (war-size exponent $\alpha \approx 1.5$; Richardson, 1948, Clauset et al., 2009).

The 33 key parameters carry literature-grounded priors; the full registry of 287 parameters with calibration-status tags, all data-source manifests, and every harness cited in the text are in the repository [Sotiriadi and Osipov, 2026].

Parameter	Central	Range	Source
Equilibrium climate sensitivity, °C	3.0	1.5–6.0	IPCC AR6 WG1; Otto et al. 2013
Surface heat capacity	8.0	6.0–18.0	Geoffroy et al. 2013
Ocean heat exchange	1.0	0.4–1.3	Geoffroy et al. 2013
Emissions scale	0.98	0.90–1.05	Global Carbon Project 2023
Structural decarbonization rate	0.016	0.010–0.030	GCP/WDI trend; 2015–2023 backtest
Capital share α	0.30	0.22–0.40	PWT 10; Gollin 2002
Labor share β	0.60	0.50–0.72	PWT 10; Gollin 2002
Energy elasticity γ	0.042	0.015–0.080	2015–2023 backtest
Capital–energy substitution σ_{KE}	0.40	0.20–0.60	van der Werf 2008; Koetse et al. 2008
Depreciation rate δ	0.05	0.03–0.08	PWT 8/9
TFP sensitivity to R&D	0.30	0.10–0.60	calibration to backtest
Damage coefficient	0.0078	0.0015–0.025	Howard & Sterner 2017; DICE-2016R2; Burke et al. 2015
Pure time preference ρ	0.015	0.001–0.030	DICE-2016R2; Stern 2007
Elasticity of marginal utility η	1.45	1.0–2.0	DICE-2016R2; Stern 2007
Market demand elasticity	0.40	0.20–0.70	Labandeira et al. 2017
<i>Social block (validated by sign and channel, Sec. 6)</i>			
Tension stress from unemployment	0.010	0.007–0.014	Di Tella et al. 2001; structural prior
Tension stress from inflation	0.005	0.003–0.007	Di Tella et al. 2001 (aversion $\approx 2:1$); prior
Inequality effect on tension	0.0005	0.0003–0.0007	Wilkinson & Pickett 2009
Trust-anchor strength	0.060	0.039–0.081	calibrated to WGI trust persistence

B. Integration benchmark: where sectoral models stop

For three shocks we take a published conclusion from a named sectoral model, feed GIM the same exogenous shock, and ask whether it (a) reproduces the sectoral first-order effect and (b) surfaces a cross-sector consequence the sectoral model cannot represent by construction. Runs are deterministic (57 countries, ten years, extreme events off); each scenario carries a dose-response sweep on which the sectoral model's slope is, by construction, zero.

Carbon tax (anchor: DICE Nordhaus, 2017). Fed the same \$50/tCO₂ tax, GIM reproduces the emissions decline ($\approx 5.5\%$ over ten years) and adds the political-economy channel: tension rises and trust falls monotonically in the rate — the mechanism behind the French 2018 and Australian 2014 reversals; DICE's response on that axis is zero.

Oil shock (anchor: energy-system models). A 20% supply cut propagates into sovereign finance: two additional importers enter debt crisis at a realistic burden, four more with triple exposure at a severe one; the response is threshold-stepped (Sri Lanka 2022 is the real-world instance). Energy-system models produce no sovereign crises at any magnitude.

Crop shock (anchor: AgMIP/AR6). A 50% regional yield loss raises food-affordability stress by ≈ 0.11 and protest pressure by ≈ 0.016 on the vulnerable subset, convex in loss depth [Lagi et al., 2011]; the assessment chain stops at the food-security index.

Across the three channels GIM shows a positive, measurable slope where every sectoral model is flat at zero (Fig. 6) — the integration dividend, quantified.

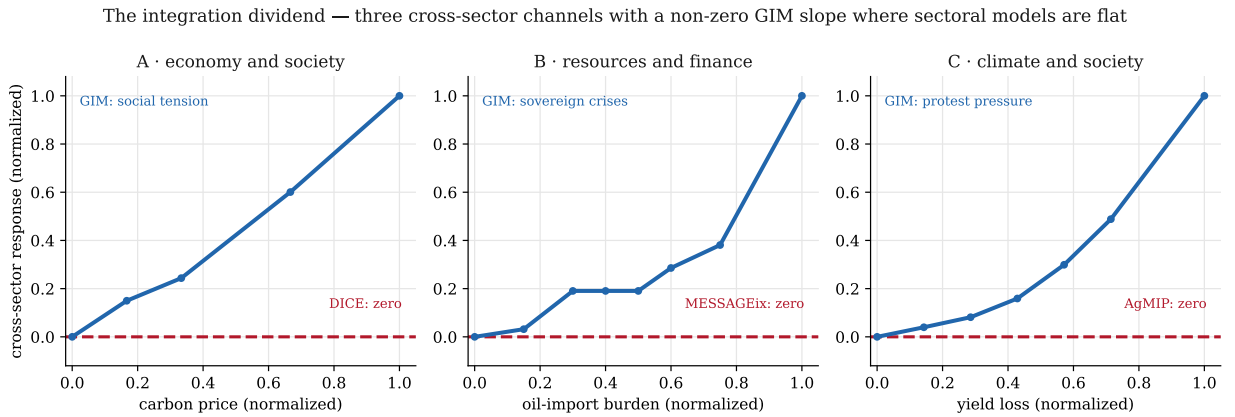


Figure 6: The integration dividend: three cross-sector channels, each normalized to its own maximum; the dashed line is every sectoral model's zero response. Smooth for economy–society, threshold-stepped for resources–finance, convex for climate–society.

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